

Memory Forensics Investigation: Process Hollowing Detection

Overview

This investigation focused on analyzing a memory image to identify evidence of malware execution, process injection, command-and-control (C2) communication, and attacker techniques. The objective was to determine:

1. The compromised process
2. The parent-child process relationship
3. Evidence of code injection
4. Network communication with the attacker's infrastructure

Tools and Techniques

1. Memory Forensics Analysis
2. Process Tree Analysis
3. Network Connection Analysis
4. Memory Injection Detection
5. Malware Behavioral Analysis

Compromised Process Identification

The investigation identified the following compromised process:

- Process Name: svchost.exe
- Process ID (PID): 3880

The process was found to contain suspicious injected code and active network communications, indicating it had been used as a vessel for malicious activity.

Parent-Child Process Analysis

- Parent Process Name: notepad.exe
- Parent Process ID (PID): 2550

Why This Is Suspicious

Windows processes typically follow a predictable hierarchy. Under normal circumstances, svchost.exe is launched by services.exe and is responsible for hosting legitimate Windows services.

In this investigation, however, svchost.exe (PID 3880) was spawned by notepad.exe (PID 2550).

This relationship is highly abnormal because Notepad is a simple user application and has no legitimate role in creating or managing service host processes.

The unexpected parent-child relationship suggests that the attacker manipulated a legitimate process to conceal malicious activity and evade detection.

Memory Injection Evidence

Memory analysis revealed clear indicators of code injection.

Suspicious Memory Protection

PAGE_EXECUTE_READWRITE

This memory protection flag allows a memory region to be:

1. Written to
2. Read from
3. Executed

Legitimate applications rarely require memory to be both writable and executable simultaneously. This configuration is frequently associated with malware injection techniques.

PE Header Detected in Memory

The presence of the MZ magic bytes indicates that a Windows Portable Executable (PE) file was loaded directly into process memory.

This suggests that executable code was injected and executed in memory rather than being launched from disk.

The combination of:

- PAGE_EXECUTE_READWRITE memory permissions
- MZ executable headers in memory

provides strong evidence of in-memory code injection and fileless malware execution.

Command-and-Control (C2) Communication

The compromised process was observed communicating with an external system.

Destination IP: 185.112.55.20

Destination Port: 443

The use of TCP port 443 may help malicious traffic blend in with normal HTTPS communications, making detection more difficult.

Attack Classification

Identified Technique: Process Hollowing

MITRE ATT&CK: T1055.012

The following findings support the Process Hollowing assessment:

1. An abnormal process hierarchy where notepad.exe spawned svchost.exe.
2. Executable code present in writable and executable memory regions.
3. Detection of PE headers (MZ) within process memory.
4. Active outbound communication to an external IP address.
5. Execution occurring primarily in memory rather than from a file on disk.

Attack Flow

1. A legitimate process (svchost.exe) was created.
2. Its memory space was modified and populated with malicious code.
3. The injected payload executed from memory.
4. The compromised process established communication with an external command-and-control server.

Skills Demonstrated

- Memory Forensics
- Malware Analysis
- Process Tree Analysis
- Threat Hunting
- Network Forensics
- MITRE ATT&CK Mapping
- Detection Engineering
- Incident Response